

# StartupET Platform

## Mobile Software Requirements Specification (SRS)

|                  |                                                   |           |                                              |
|------------------|---------------------------------------------------|-----------|----------------------------------------------|
| Document Ref:    | STARTUPET-SRS-MOB-2026-V1.0                       | Standard: | IEEE Std 29148-2018 / IEEE Std 830-1998      |
| Target Platform: | Flutter Cross-Platform (iOS 12.0+ / Android 5.0+) | Status:   | Production Standard / Approved Specification |

## Software Requirements Specification (SRS)

### StartupET Cross-Platform Mobile Application

**Document Reference:** STARTUPET-SRS-MOB-2026-V1.0

**Standard Compliance:** IEEE Std 29148-2018 / IEEE Std 830-1998

**Classification:** Technical Specification / Production Standard

**Target Platform:** Flutter Mobile Engine (iOS 12.0+ & Android 5.0+ / API Level 21+)

**Backend System:** Next.js 15 App Router API Subsystem

**Date:** August 10, 2026

**Document Status:** Approved Specification

### Document Control & Revision History

| Version | Date            | Author / Role                | Description of Changes                                       | Approved By         |
|---------|-----------------|------------------------------|--------------------------------------------------------------|---------------------|
| v0.1    | July 15, 2026   | Mobile Engineering Team      | Initial Requirements Draft & Domain Exploration              | Lead Architect      |
| v0.5    | July 28, 2026   | Security & API Working Group | Added Fayda OIDC & Secure Storage Requirements               | Security Officer    |
| v1.0    | August 10, 2026 | Principal System Architect   | Finalized IEEE 29148 SRS Specification & Traceability Matrix | Head of Engineering |

## 1. Introduction

### 1.1 Purpose

This document provides the formal **Software Requirements Specification (SRS)** for the **StartupET Mobile Client**. It details the functional, non-functional, interface, security, and data handling requirements for the native mobile application on iOS and Android platforms.

This specification serves as the authoritative technical contract between product management, UI/UX designers, mobile engineering teams, backend integration developers, and quality assurance testers.

1.2 Scope of the Application

The StartupET Mobile Application provides an integrated mobile gateway for Ethiopian startup founders, ecosystem builders (incubators, accelerators, investors), and program applicants.

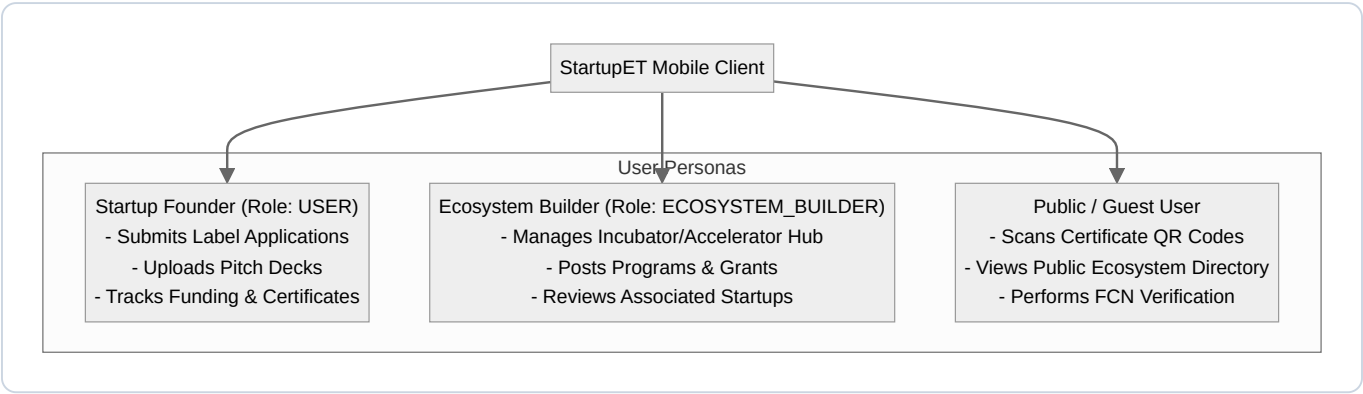
Key product capabilities encompassed by this SRS include:

- Mobile authentication, credentials exchange, and session persistence.
- Identity verification via Ethiopia's **Fayda National ID System** (16-digit FCN OIDC integration) and business tax TIN verification.
- Startup registration, profile editing, team founder management, and pitch deck document attachment.
- End-to-end startup label certification application lifecycle management (Drafting, Submission, Under Review, Approval, Rejection).
- Digital Startup Label Certificate display, PDF downloading, and QR-code authenticity validation.
- Ecosystem program directory, grant/funding opportunity search, and application tracking.
- Pitch deck management and mentor rating/feedback review.
- Real-time push alerts and in-app notification center.

1.3 Definitions, Acronyms, and Abbreviations

| Term / Acronym | Full Definition                                                               |
|----------------|-------------------------------------------------------------------------------|
| SRS            | Software Requirements Specification (IEEE 29148 Standard)                     |
| FCN            | Fayda Code Number (16-digit Ethiopian National Identity Number)               |
| OIDC           | OpenID Connect (Authentication Protocol built on OAuth 2.0)                   |
| BLoC           | Business Logic Component (Flutter State Management Architecture Pattern)      |
| JWT            | JSON Web Token (Session Token Format)                                         |
| TIN            | Tax Identification Number (Ethiopian Ministry of Revenue Business Identifier) |
| RBAC           | Role-Based Access Control                                                     |
| SSE            | Server-Sent Events (Real-time Notification Streaming)                         |
| Dio            | Configurable HTTP Client for Dart/Flutter                                     |
| DI             | Dependency Injection (Implemented via <code>GetIt</code> package)             |

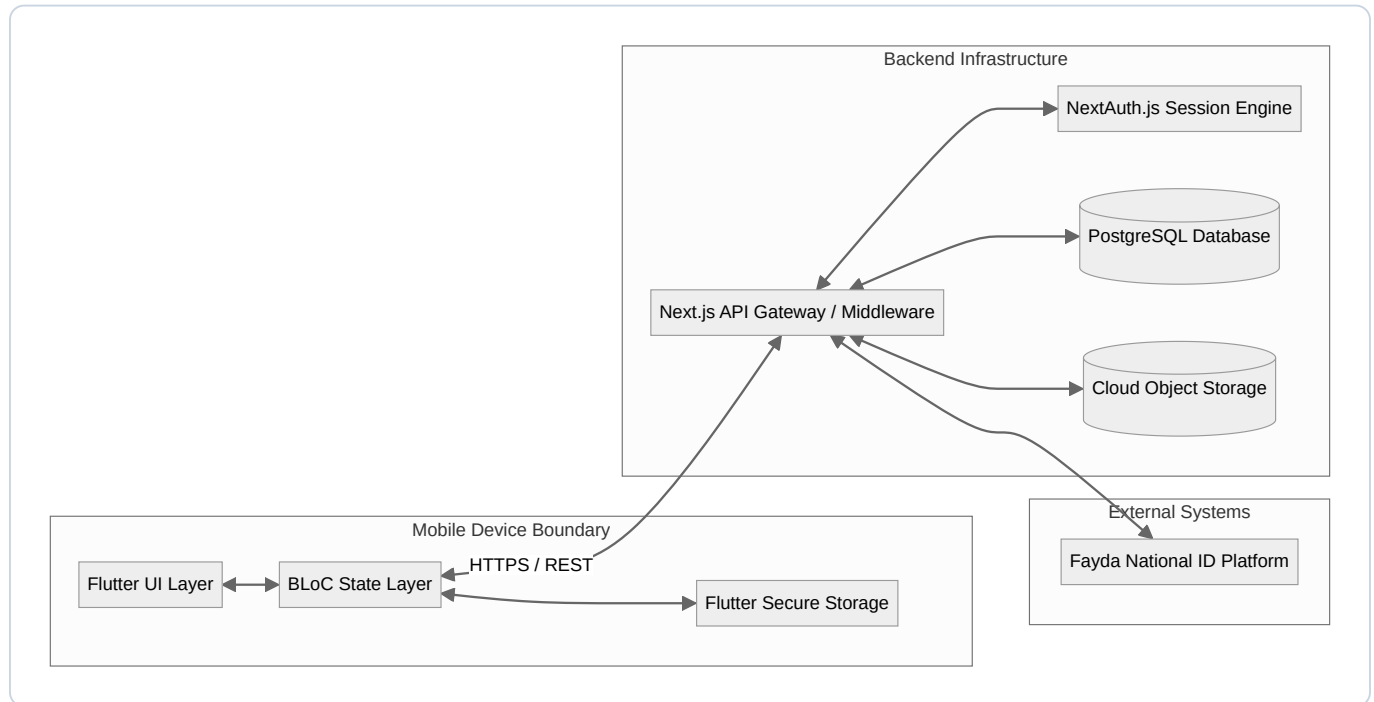
1.4 User Classes & Operational Personas



## 2. Overall Description

### 2.1 Product Perspective

The mobile client operates as a decoupled front-end interfacing with the central Next.js 15 App Router backend server over HTTPS RESTful web services.



### 2.2 Product Functional Summary

The application is organized into six core functional modules:

1. **Authentication & Identity Engine:** Handles login, JWT token caching, session auto-revalidation, logout, and Fayda National ID verification.
2. **Startup Profile & Business Verification:** Manages company metadata, founder equity distributions, and TIN verification.
3. **Application & Certificate Engine:** Form submission, document attachments, review stage tracking, and QR certificate rendering.
4. **Ecosystem & Funding Module:** Browsing grant programs, incubator hubs, and funding applications.
5. **Pitch Deck & Feedback Engine:** Pitch deck uploads, version management, and score/feedback display.
6. **Notification System:** Server-Sent Events (SSE) background stream, unread badge counts, and state transition alerts.

### 2.3 Operating Environment & Constraints

- **Mobile Operating Systems:** iOS 12.0 or higher; Android 5.0 (API Level 21) or higher.
- **Framework:** Flutter 3.x with Dart 3.x.
- **Network Requirements:** Active cellular (3G/4G/5G) or Wi-Fi connectivity. Graceful offline mock handling via `MockInterceptor`.
- **Storage Constraints:** App footprint < 60 MB; Secure keychain storage for tokens.

## 3. Detailed Functional Requirements

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### 3.1 Module 1: Authentication & Identity Management

#### FR-AUTH-001: Credentials Login

- **Description:** The mobile app shall allow users to log in using an email address and password via `POST /api/auth/callback/credentials`.
- **Pre-conditions:** User account must exist in the database and not be suspended.
- **Inputs:** `email` (string, valid email format), `password` (string, minimum 8 characters).
- **Processing:**
  1. Mobile app validates input fields locally.
  2. Mobile app fetches CSRF token from `GET /api/auth/csrf` (if required by web container) or posts directly.
  3. Sends request via `ApiClient`.
  4. On HTTP 200, extracts `sessionToken` / JWT token from response payload and cookies.
  5. Saves token to `FlutterSecureStorage` under key `api_key`.
  6. Emits `AuthenticatedState` in `AuthBloc`.
- **Outputs:** User object containing `id`, `email`, `role`, `firstName`, `lastName`, and session token.
- **Error Handling:**
  - HTTP 401 ("Invalid email or password"): Surface user-friendly error snackbar.
  - HTTP 429 ("Too many failed login attempts"): Lock screen for 15 minutes.

#### FR-AUTH-002: Secure Token & Session Persistence

- **Description:** The app shall automatically attach the stored session token as a Bearer header ( `Authorization: Bearer <token>` ) and as a `next-auth.session-token` cookie header on every outbound Dio request.
- **Verification:** Unit & Integration tests verifying `_authInterceptor` in `ApiClient`.

#### FR-AUTH-003: Fayda National ID (16-Digit FCN) Verification

- **Description:** The application shall allow authenticated users to verify their Ethiopian National Identity by submitting their 16-digit Fayda Code Number (FCN) to `GET /api/verify-national-id?fcn=<16-digit-number>`.
- **Pre-conditions:** User must be authenticated ( `AuthBloc` in `AuthenticatedState` ).
- **Input Validation:** FCN string must strictly match regex `^\d{16}$`.
- **Processing:**
  1. API Client issues request with rate-limiting guard (max 20 requests/hour/user).
  2. Server queries Fayda OIDC platform.
  3. App parses response payload into `VerificationRecord`.
- **Outputs:** Verification status ( `VERIFIED`, `NOT_FOUND`, `UNAVAILABLE` ), demographic verification flags.

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### 3.2 Module 2: Startup Profile & Business Verification

#### FR-PROF-001: Startup Profile Management

- **Description:** Startup founders shall be able to view and edit their company profile including company name, sector, stage ( `INITIAL`, `POST_INITIAL`, `SCALE_EXIT` ), employee count, capital, and founder equity allocation.
- **Input Validation:** Total share percentages across all listed co-founders must sum to at least 25% of total company shares.

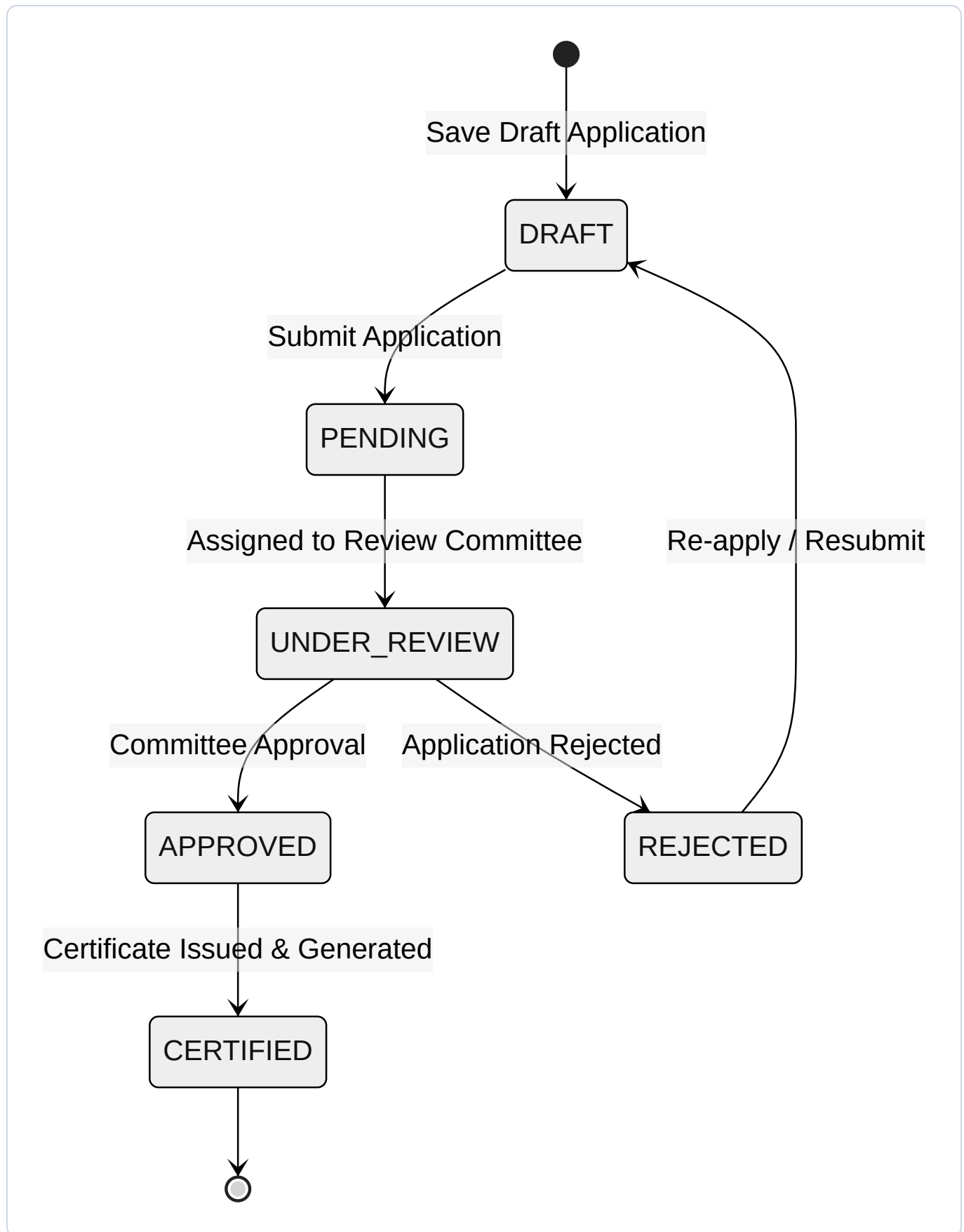
#### FR-PROF-002: Business Tax ID (TIN) Verification

- **Description:** The app shall verify business legitimacy by querying `GET /api/verify-tin?tin=<tin_number>` which validates the TIN against government e-trade registry records.
  - **Outputs:** `businessName` , `tinNumber` , `registrationNumber` , `status` .
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### 3.3 Module 3: Labeling & Certificate Application Engine

#### FR-APP-001: Startup Label Application Submission

- **Description:** Founders shall be able to submit a formal application for official Ethiopian Startup Label Certification via `POST /api/applications` .
- **Required Input Fields:**
  - `startupName` (string)
  - `industry` (string)
  - `stage` ( `INITIAL` | `POST_INITIAL` | `SCALE_EXIT` )
  - `businessEmail` (valid email)
  - `phoneNumber` (string)
  - `totalShares` (numeric string)
  - `sharePrice` (numeric string)
  - `founders` (array of co-founders with 16-digit FCN)
  - `businessLicenseFileId` , `articlesOfIncorpFileId` , `pitchDeckFileId` (valid file IDs)
  - `status` : "PENDING"
- **Business Rule:** A startup founder can only have **one active in-flight application** at a time. Attempts to post a second active application shall return `HTTP 409 Conflict` .

**FR-APP-002: Startup Label Certificate & QR Code Verification**

- **Description:** Upon application approval, the app shall fetch the issued certificate via `GET /api/startups/my-status` and allow downloading/rendering the high-resolution PDF certificate with embedded QR verification code via `GET /api/certificates/[id]/download`.

- **Inputs:** Certificate ID or Application ID.
  - **Outputs:** Native PDF viewer rendering or saving to device storage.
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### 3.4 Module 4: Ecosystem & Funding Opportunities

#### FR-FUND-001: Grant & Opportunity Directory

- **Description:** The app shall display available grants, incubator acceleration programs, and funding opportunities posted by registered Ecosystem Builders.
- **Features:** Search by keyword, filter by grant amount, filter by deadline date.

#### FR-FUND-002: Funding Application Submission

- **Description:** Founders shall submit applications to specific funding programs, tracking real-time status from submission through disbursement.
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### 3.5 Module 5: Pitch Deck Management & Feedback

#### FR-PITCH-001: Pitch Deck Upload & Version Control

- **Description:** Founders can upload PDF pitch decks using multipart upload ( `POST /api/files/upload` ), attaching the returned `fileId` to their startup profile.
- **File Constraints:** Maximum file size 10 MB; accepted MIME types `application/pdf`.

#### FR-PITCH-002: Reviewer Rating & Feedback Display

- **Description:** Founders can view scoring metrics (Problem, Solution, Market, Innovation, Scalability) and qualitative comments provided by committee reviewers.
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### 3.6 Module 6: Notification & Alert Engine

#### FR-NOTIF-001: Server-Sent Events (SSE) Stream

- **Description:** The mobile app shall maintain an active SSE listener on `GET /api/notifications/stream` while in the foreground to receive instant application status changes.

#### FR-NOTIF-002: Local Push Alert Dispatching

- **Description:** When an SSE notification event arrives (e.g., `CERTIFICATE_ISSUED` or `APPLICATION_REJECTED` ), the app shall issue a system notification drawer alert.
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## 4. External Interface Requirements

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### 4.1 User Interface (UI) Requirements

- **Design System:** Modern, high-contrast dark/light mode tailored palette using standard Material 3 / Cupertino design principles.
- **Responsiveness:** Support screen sizes from 4.7" (iPhone SE) to 6.7" (Pro Max) and Android tablets.
- **Language Support:** English and Amharic localized string resources.

### 4.2 Software & Network Interfaces

- **Transport Protocol:** HTTPS over TLS 1.3.

- **Data Serialization:** JSON format for API request/response bodies.
- **File Uploads:** `multipart/form-data` for document attachments.

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## 5. Non-Functional Requirements (NFRs)

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### 5.1 Performance Requirements

- **NFR-PERF-001:** Mobile application cold start time shall be less than **2.0 seconds** on standard mid-range hardware.
- **NFR-PERF-002:** UI rendering frame rate shall maintain **60 FPS** without frame drops during scroll and transition animations.
- **NFR-PERF-003:** API Client connect timeout shall be set to **15,000 ms** and receive timeout to **15,000 ms**.

### 5.2 Security & Confidentiality Requirements

- **NFR-SEC-001:** Auth tokens and sensitive user payload shall be encrypted at rest using OS keychain (iOS `KeychainManager`, Android `EncryptedSharedPreferences` via `FlutterSecureStorage` ).
- **NFR-SEC-002:** Communication channels must enforce TLS 1.3 encryption.
- **NFR-SEC-003:** Session tokens shall automatically expire after **72 hours** of inactivity.

### 5.3 Reliability & Availability

- **NFR-REL-001:** App crash rate shall not exceed **0.1%** per active user session.
- **NFR-REL-002:** Network errors must trigger automatic retry mechanisms with exponential backoff for read operations.

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## 6. Requirements Traceability Matrix (RTM)

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The matrix below maps each Functional Requirement to its corresponding Flutter BLoC component, Service implementation, and Backend API endpoint:



| Requirement ID | Requirement Description | Mobile BLoC / Component | Mobile Service Class | Target Backend API Endpoint         |
|----------------|-------------------------|-------------------------|----------------------|-------------------------------------|
| FR-AUTH-001    | Credentials Login       | AuthBloc                | AuthService          | POST /api/auth/callback/credentials |
| FR-AUTH-002    | Session Persistence     | ApiClient               | FlutterSecureStorage | All Endpoint Requests ( Header )    |
| FR-AUTH-003    | Fayda FCN Verification  | VerificationBloc        | VerificationService  | GET /api/verify-national-id         |
| FR-PROF-001    | Profile Management      | StartupBloc             | StartupService       | GET /api/startups/my-status         |
| FR-PROF-002    | Business TIN Check      | VerificationBloc        | VerificationService  | GET /api/verify-tin                 |
| FR-APP-001     | Label App Submission    | ApplicationBloc         | ApplicationService   | POST /api/applications              |
| FR-APP-002     | Certificate Download    | CertificateBloc         | CertificateService   | GET /api/certificates/[id]/download |
| FR-FUND-001    | Browse Funding Grants   | FundingBloc             | FundingService       | GET /api/funding                    |
| FR-PITCH-001   | Pitch Deck Attachment   | PitchBloc , FileBloc    | FileService          | POST /api/files/upload              |
| FR-PITCH-002   | Reviewer Rating View    | PitchBloc               | PitchService         | GET /api/applications/[id]          |
| FR-NOTIF-001   | Real-time Notification  | NotificationBloc        | NotificationService  | GET /api/notifications/stream       |

### Sign-off & Formal Approval:

- **Product Manager:** *Approved*
- **Mobile Engineering Lead:** *Approved*
- **Lead QA Engineer:** *Approved*